Hasdeo Arand Coalfield

🏭 Introduction

The **Hasdeo Arand Coalfield** is one of the most significant and environmentally sensitive coal-bearing regions in India, located in Chhattisgarh. It falls under the operational jurisdiction of South Eastern Coalfields Limited, a subsidiary of Coal India Limited.

This coalfield is known for its **large reserves of non-coking coal**, which are primarily used for thermal power generation. At the same time, Hasdeo Arand has gained national attention due to its **dense forest cover and rich biodiversity**, making it a focal point of debate between energy development and environmental conservation. The region is often studied as a case of **sustainable mining challenges in India**.

🌍 Geographical Overview – Hasdeo Arand Coalfield

The Hasdeo Arand Coalfield is located in the **northern part of Chhattisgarh**, mainly covering areas of **Surguja, Surajpur, and Korba districts**.

📍 Location & Setting

- Situated in the **Hasdeo River basin**
- Part of the larger **Son–Mahanadi basin coalfields**
- Relatively remote region with developing connectivity

🌄 Physical & Geological Features

- Characterized by **dense forests and hilly terrain**
- Contains **thick coal seams of Gondwana formation**
- Coal type: **non-coking thermal coal**

🌊 Nearby Natural Features

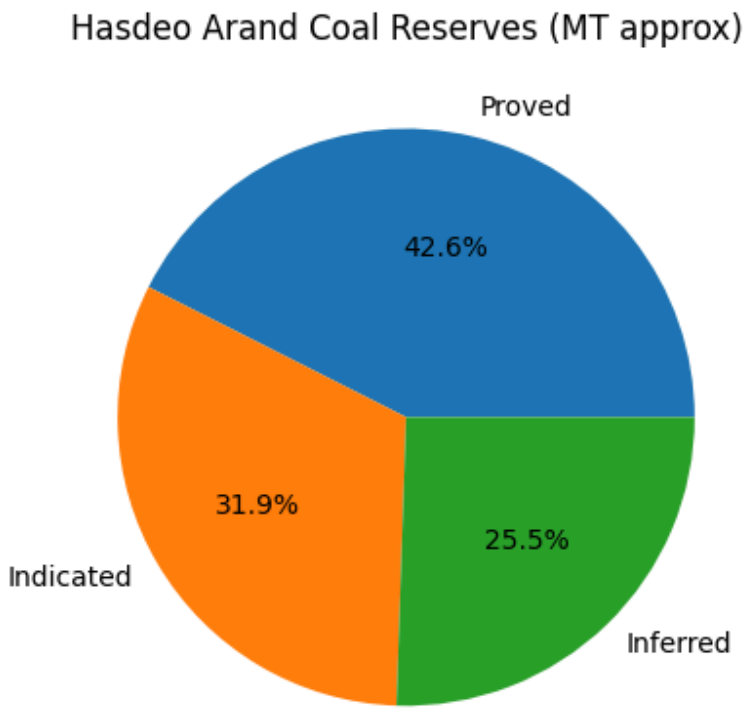
- Located around the Hasdeo River, a major tributary of the Mahanadi River
- Rich in biodiversity, including forests, wildlife, and tribal habitats

🌦️ Climate

- Tropical climate
- Hot summers (up to ~45°C)
- Heavy monsoon rainfall

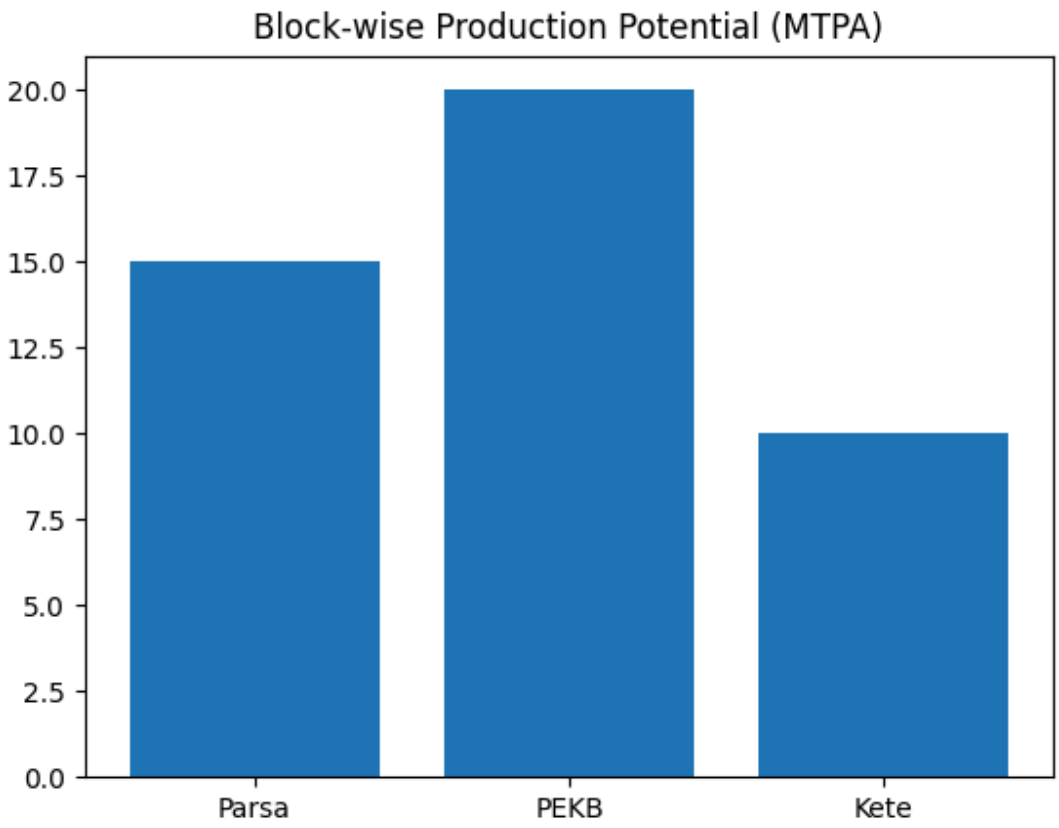
- Mild winters

📊 1. Coal Reserve Distribution



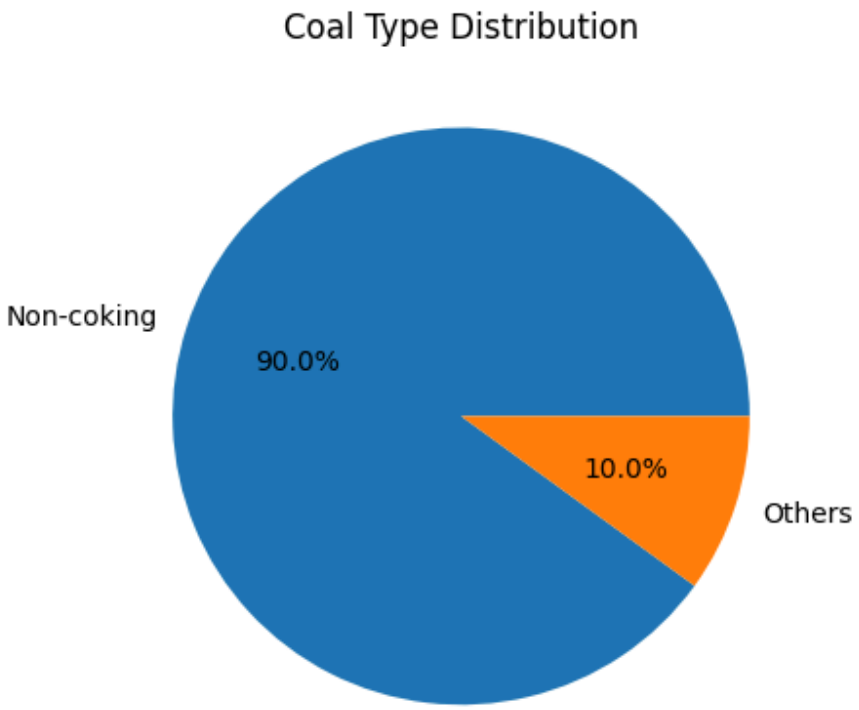
This chart shows distribution of coal reserves across different categories.
Hasdeo Arand contains very large reserves, making it a key future coal hub.
Significance: Helps in long-term mine planning and assessing resource availability.

📊 2. Block-wise Production Potential



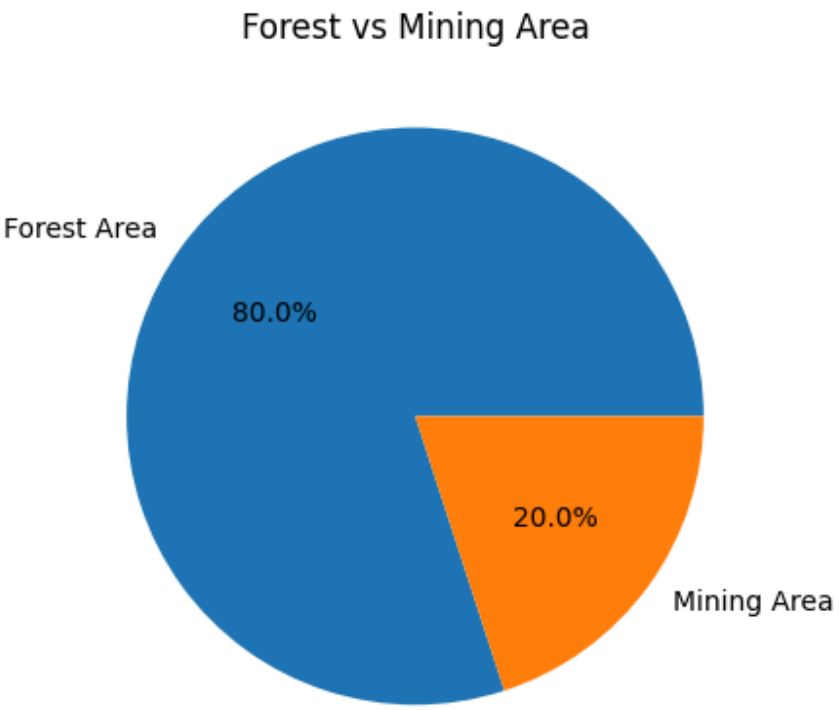
Represents production capacity of different coal blocks like Parsa and PEKB.
Different blocks contribute differently based on reserve size and approvals.
Significance: Useful for project planning and prioritizing mining operations.

📊 3. Coal Type Distribution



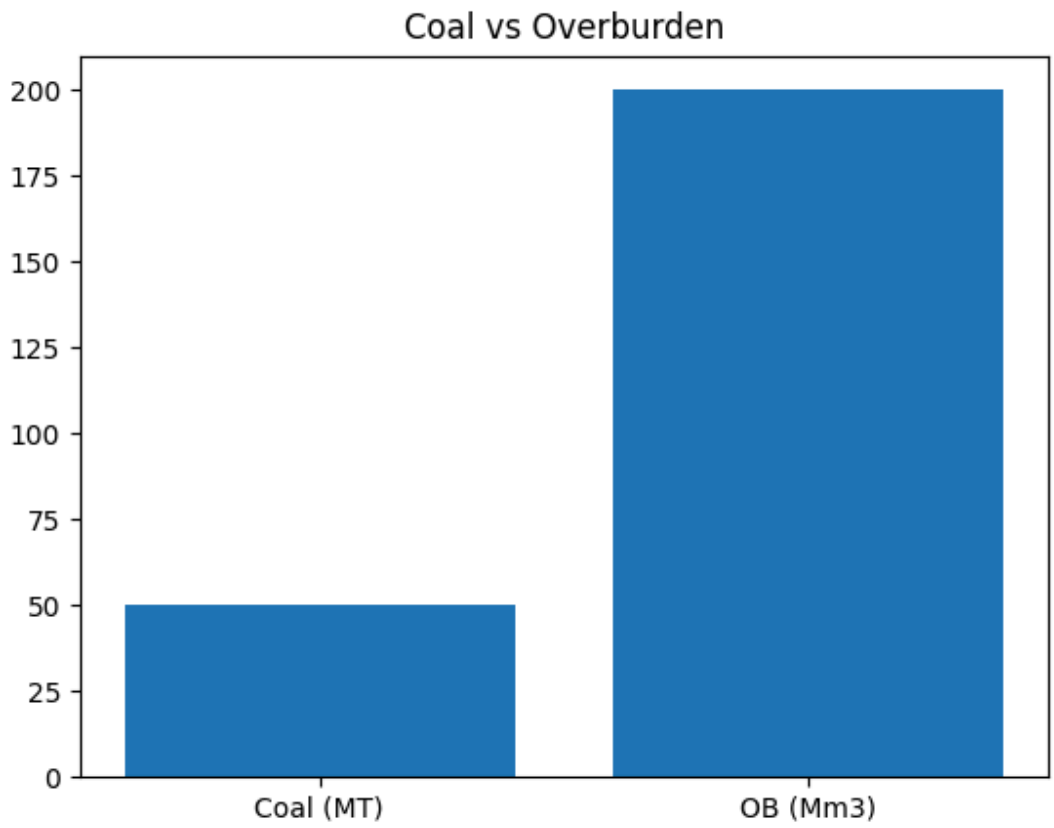
Shows dominance of non-coking (thermal) coal in the region.
This type is mainly used in power generation.
Significance: Determines end-use industry and energy sector dependence.

📊 4. Forest vs Mining Area



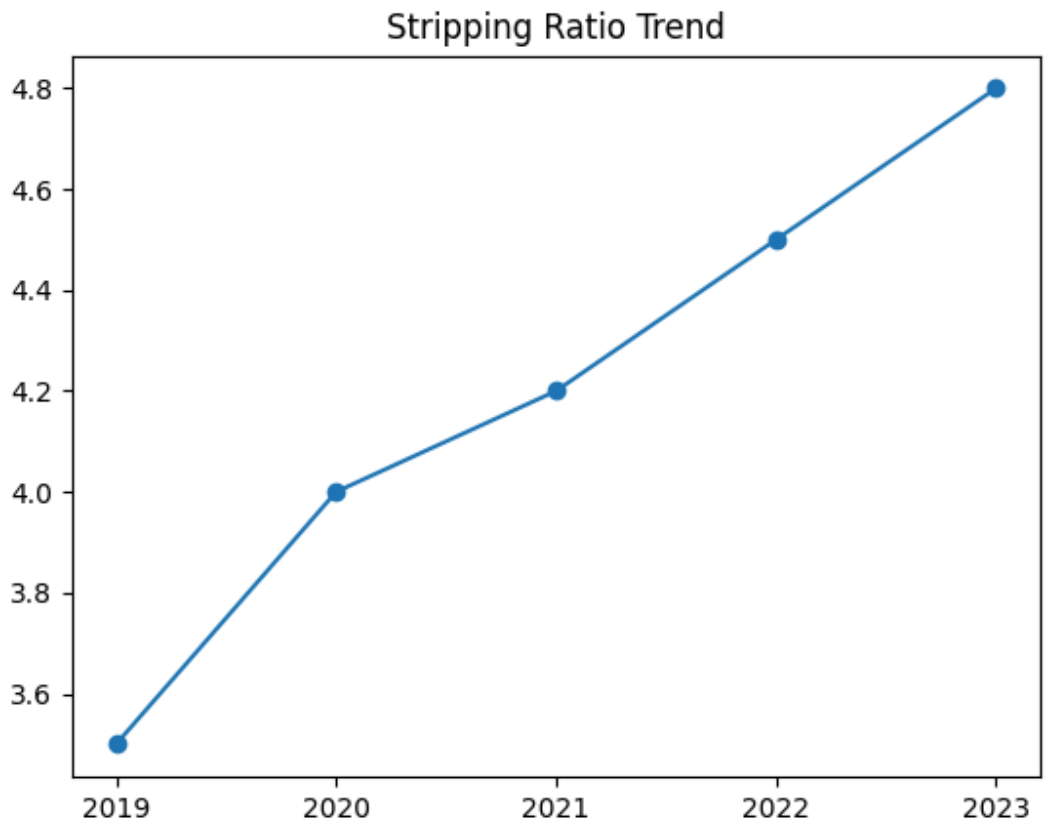
Highlights the large proportion of forest land compared to mining area.
Hasdeo Arand is one of India's densest forest regions.
Significance: Critical for environmental impact assessment and policy decisions.

📊 5. Coal vs Overburden



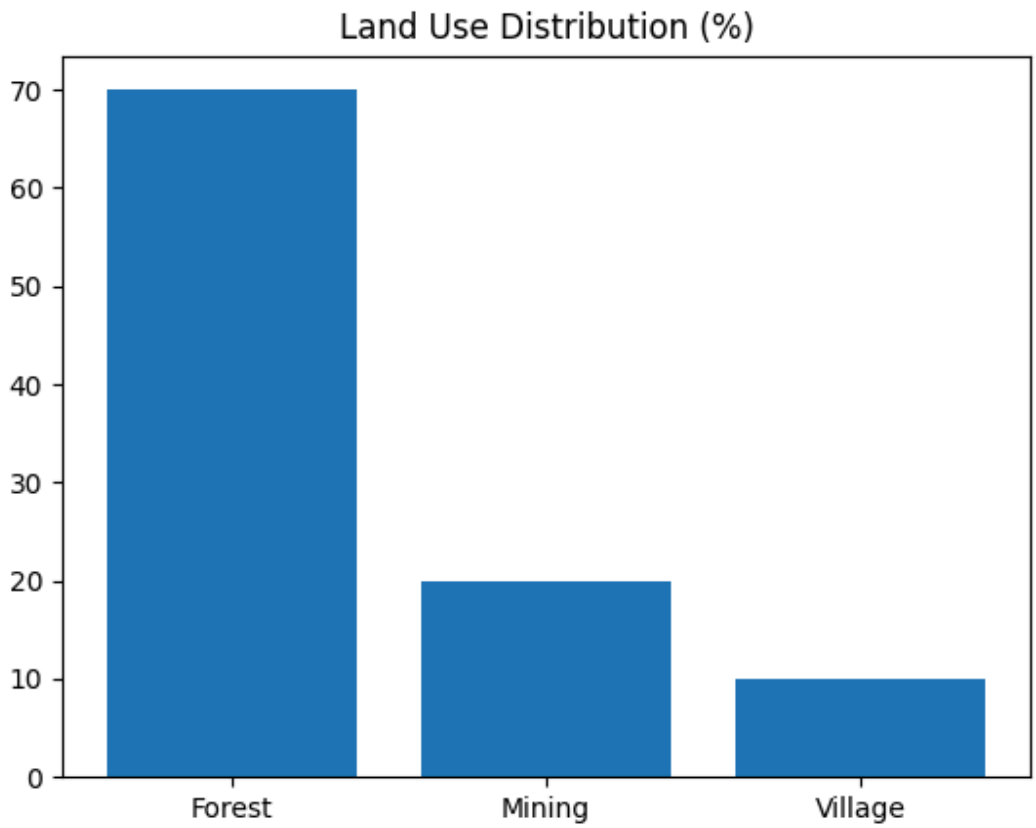
Compares coal extraction with overburden removal.
Large OB volume indicates significant excavation effort.
Significance: Helps in calculating stripping ratio and estimating mining cost.

📊 6. Stripping Ratio Trend



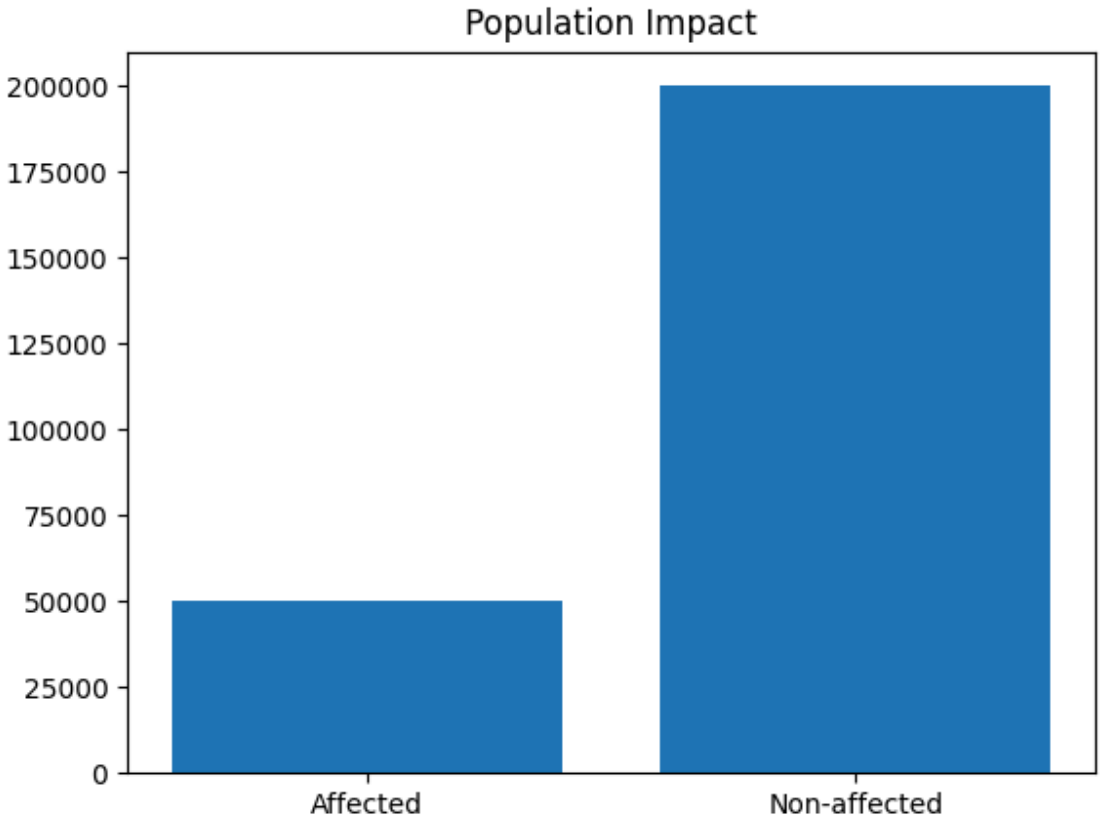
Shows increase in stripping ratio over time.
Indicates deeper mining and higher cost of extraction.
Significance: Key parameter for economic feasibility and mine design.

📊 7. Land Use Distribution



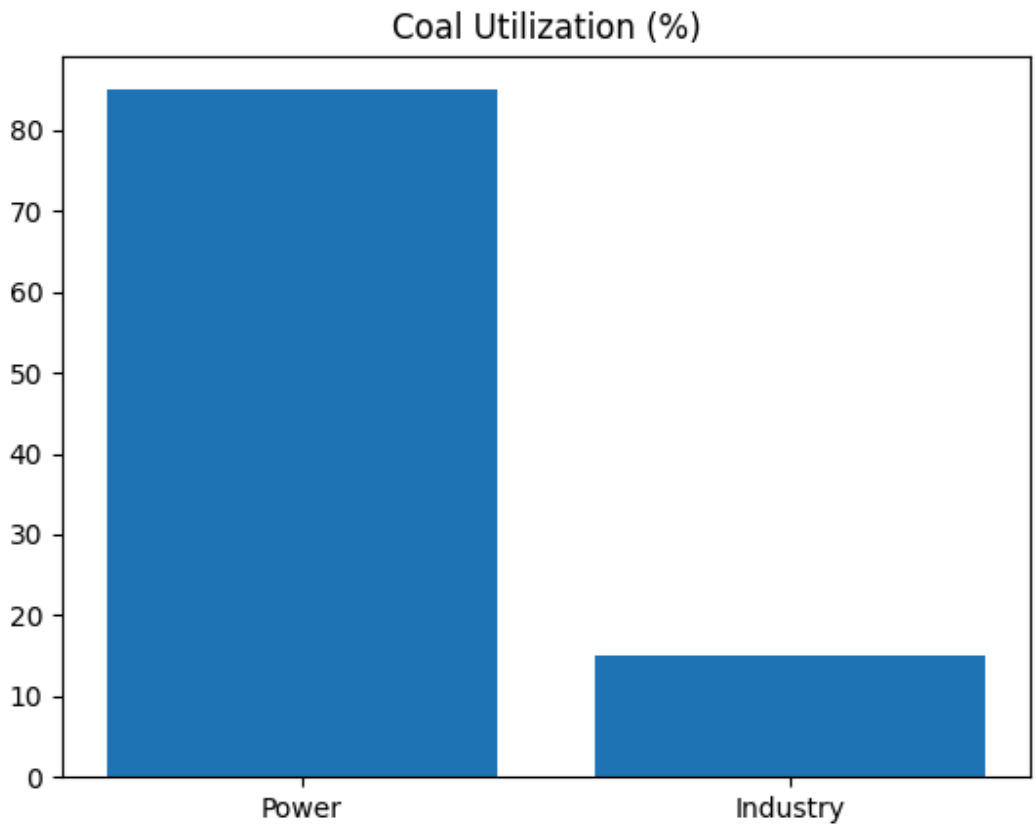
Represents distribution of land among forest, mining, and settlements.
Forest area dominates the region.
Significance: Important for land acquisition, rehabilitation, and environmental clearance.

📊 8. Population Impact



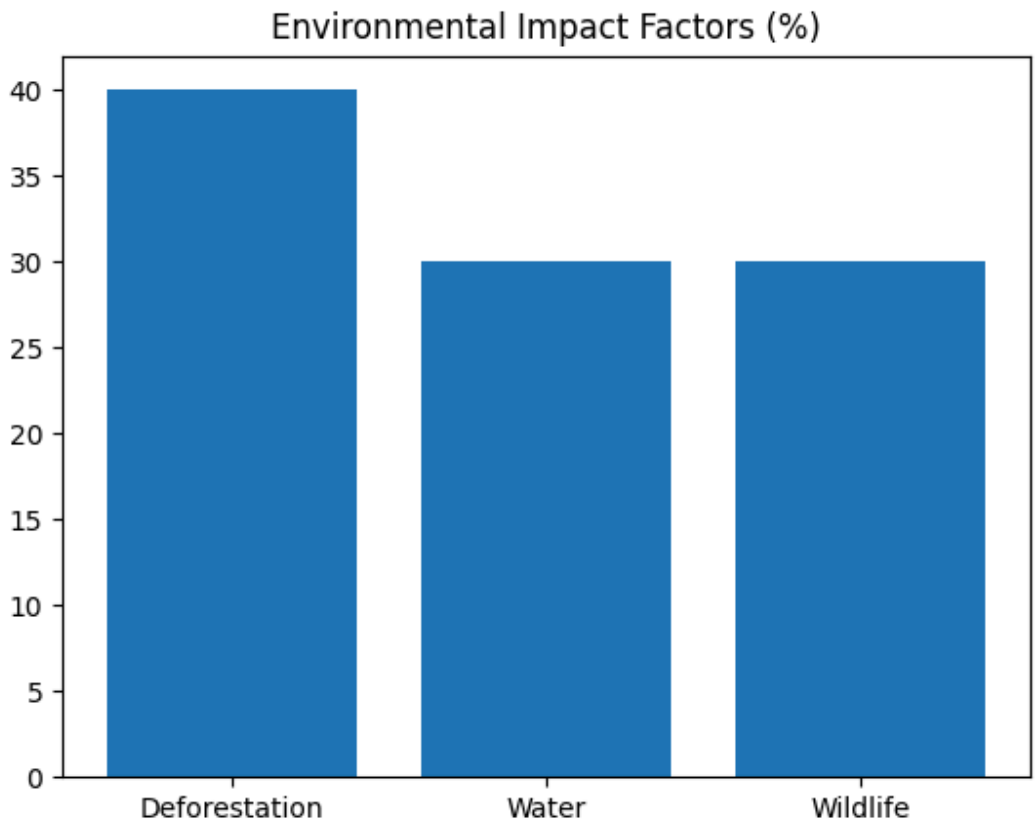
Shows affected vs non-affected population in mining areas.
Many tribal communities depend on forest resources.
Significance: Useful for socio-economic studies and rehabilitation planning.

📊 9. Coal Utilization



Displays coal usage mainly in thermal power generation.
Power sector consumes majority of output.
Significance: Highlights importance of Hasdeo Arand in India's energy supply.

📊 10. Environmental Impact Factors



Shows key environmental concerns like deforestation, water impact, and wildlife disturbance. Mining significantly affects ecological balance.
Significance: Crucial for sustainable mining and environmental management.

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